Stefan M. Schulz

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EDUCATION

University of Bonn 12/2024 – 12/2027 (Expected)
PhD in Computer Science (Differentiable Scene Representation for 4D/6D Capture, Reconstruction, and Synthesis) Bonn, Germany

- Advised by Prof. Matthias Hullin & Prof. Reinhard Klein; researching efficient real-time novel view synthesis, 4D scene/video compression, and multi-modal reconstruction.

University of Bonn 04/2023 – 10/2024
Master of Science in Computer Science | Grade: 1.4 Bonn, Germany

- Completed semester abroad at **Aalto University, Finland** (08/2023 – 01/2024).
- Thesis adapted into a second-author ICLR conference paper.
- Deutschlandstipendium 2023-2024.

University of Bonn 10/2019 – 03/2023
Bachelor of Science in Computer Science | Grade: 1.3 (Excellent) Bonn, Germany

- Deutschlandstipendium 2021-2023.

RESEARCH & INDUSTRY EXPERIENCE

University of Bonn 12/2024 – Present
PhD Research Staff Bonn, Germany

- Co-engineered the physical assembly and software infrastructure for a state-of-the-art, multi-camera light stage.
- Developed an interactive Python/PyTorch diagnostic module for visualization and multi-view perspective switching.
- Programmed procedural Blender pipelines to render accurate synthetic simulation datasets mimicking the capture setup.

University of Bonn 08/2021 – 10/2024
Research Assistant (Computer Vision & Robotics) Bonn, Germany

- **Robotics (NimbRo Team):** Contributed to the hardware construction and software integration for autonomous humanoid platforms, ensuring optimal system stability and real-time execution; **Won the AdultSize Humanoid League World Championship**, Technical Challenges, and Best-Humanoid honors at RoboCup 2022.
- **Image Inpainting:** Designed and trained partial convolution deep networks for 3D temporal temperature satellite data reconstruction, directly resulting in a co-first-author peer-reviewed publication.
- **Medical CV:** Developed custom instance-aware deep segmentation models for automated structural analysis of medical OCT scans.

University of Bonn 09/2021 – Present
Graduate Teaching Assistant & Team Lead Bonn, Germany

- Giving lectures, managing tutorial operations, designing competitive examinations, and supervising student instructor teams for large-scale courses of up to 250 students (Technical Informatics and Computer Graphics).
- Completed teaching assistant tenures spanning Data-Centric Computer Science, Computational Intelligence, Intelligent Vision Systems, and Foundations of Robotics.

PUBLICATIONS

Peer-Reviewed Publications Accepted / Published

- **S. Schulz**, F. Edelstein, H. Dröge, R. Klein, M. Plack. "TAC-4DGS: Trajectory-Aware Compression for Rate-Distortion Optimized 4D Gaussian Splatting." *WACV*, 2027.
- N. Wandel, **S. Schulz**, R. Klein. "Metamizer: a Versatile Neural Optimizer for Fast and Accurate Physics Simulations." *ICLR*, 2025.
  - *Invited Presentation:* Presented the Metamizer project framework at the HiGraphics Workshop in April 2025 (hosted by University of Bayreuth).
- F. Huber*, **S. Schulz***, V. Steinhage. "Deep Interpolation of Remote Sensing Land Surface Temperature Data with Partial Convolutions." *MDPI Sensors*, 2024. (*Equal Contribution)
- D. Pavlichenko, ..., **S. M. Schulz**, et al. "RoboCup 2022 AdultSize Winner NimbRo." *RoboCup World Cup XXV*, Springer, 2023.

Papers Under Review & Conference Pipeline Current Submissions

- **S. Schulz***, F. Edelstein*, M. B. Hullin, R. Klein. "Aligning the Inverse: Learned Motion Compensation for Gradient-Illumination Appearance Capture." *Under Review: VISAPP 2027*. (*Equal Contribution)

- **S. Schulz**, F. Edelstein, H. Dröge, M. B. Hullin, M. Plack. “3DTV: A Feedforward Interpolation Network for Real-Time View Synthesis.” *Under Review: VISAPP 2027*.
- **S. Schulz**, F. Edelstein, H. Dröge, R. Klein, M. Plack. “Exploiting Inter-Trajectory Redundancy for Compressing 4D Gaussian Splatting.” *Under Review: VISAPP 2027*.
- F. Edelstein, **S. Schulz**, M. B. Hullin, M. Plack. “Sharper Together: Cross-Camera Feature Alignment for Multi-View Consistent Super-Resolution.” *Under Review: VISAPP 2027*.
- F. Edelstein, **S. Schulz**, R. Klein, M. Plack. CoN-SR: Consistency-Optimised Neighbourhood Super-Resolution for 3D Synthesis. *Under Review: Eurographics 2027*.

Selected Works in Progress Target Venues

- **S. Schulz**, et al. Compression method for streamable 4D Gaussian Splatting. *Target: CVPR 2027*.
- **S. Schulz**, et al. Real-time viewer for compressed 4DGS. *Target: SIGGRAPH TC 2027*.
- F. Edelstein, **S. Schulz**, et al. Water Simulation *Target: SIGGRAPH 2027*.

TECHNICAL SKILLS & CERTIFICATIONS

Core Expertise: Deep Learning, 3D/4D Scene Reconstruction & Compression, Novel View Synthesis (NeRF, Gaussian Splatting, Feedforward), Video Compression, Generative Models, Computer Vision
Programming Languages: Python, C++
Frameworks & Tools: PyTorch, git, Docker, OpenGL, Blender, LaTeX, Linux/Bash
Spoken Languages: German (Native), English (C1/Fluent), Spanish (B2), French (B1)
Technical Certifications: Deep Learning Specialization · Natural Language Processing Specialization · Machine Learning Specialization · Practical Intro to DL for CV
Professional Development & Leadership: Joint Skills Enhancement (Intercultural Awareness & Project Management) · GUSI Intercultural Training Program (Global Collaboration & Communication)